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Constraining on- and off-fault nonlinear dynamic rupture parameters via hierarchical Bayesian inversion for the 2019 Mw 7.1 ridgecrest earthquake

Zihua Niu [a], Maximilian Kruse [b], Linus Seelinger [b], Nico Schliwa [a], Heiner Igel [a], Alice-Agnes Gabriel [a,c,*]

[a] Department of Earth and Environmental Sciences, Ludwig-Maximilians-Universität München, Munich, Germany, Munich, Germany
[b] Scientific Computing Center Karlsruhe Institute of Technology, Karlsruhe, Germany
[c] Scripps Institution of Oceanography, UC San Diego, La Jolla, CA, USA

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ABSTRACT

We present a multilevel Bayesian inversion to quantify uncertainties and correlations among on- and off-fault dynamic rupture parameters for the 2019 M_w7.1 Ridgecrest earthquake. Our realistic rupture models incorporate strong velocity-weakening rate-and-state friction and off-fault plasticity. We invert for four parameters capturing along-strike variability in on-fault friction and off-fault plasticity. The inversion is constrained by multidisciplinary surface deformation data, including fault-parallel offsets from satellite imagery, high-rate GNSS time series, and static GNSS displacements, and is enabled by more than four million CPU hours of 3D dynamic rupture simulations. We find a strong correlation between on-fault frictional weakening and off-fault plasticity: increased inelastic deformation can compensate for more prominent velocity-weakening friction in the northwestern part of the fault. The preferred dynamic rupture models reduce velocity weakening from northwest to southeast along the main fault. This improves the fit with observed fault-parallel surface offsets, consistent with along-strike variations in fault maturity. The inversion favors low off-fault plastic cohesion, which improves the fit to observed surface offsets and is consistent with shallow damage zones imaged geophysically. We infer a shallow slip deficit (SSD) of 13.1% with a standard deviation of 5.1%. Notably, 55% of the models fall within the 8–32% SSD range estimated from previous kinematic inversions, implying that co-seismic off-fault inelasticity is crucial to explain the observed SSD. Our results demonstrate the feasibility of integrating 3D dynamic rupture simulations with multilevel Bayesian inversion to constrain nonlinear dynamic rupture processes and off-fault properties, toward probabilistic, physics-based earthquake source characterization and uncertainty quantification grounded in observational constraints.

1. Introduction

The M_w 7.1 Ridgecrest earthquake struck California on 5 July 2019 (Ross et al., 2019) and ruptured across a complicated multi-fault system (e.g., DuRoss et al., 2020). Multiple types of observations captured the event, including strong ground motion seismic records (Hauksson et al., 2020), global navigation satellite system (GNSS) co-seismic displacements (Melgar et al., 2020; Floyd et al., 2020), and fault displacement discontinuities (Antoine et al., 2021), hereafter referred to as "offsets", from optical image correlation using satellite images. These high-quality datasets have enhanced data-driven interpretation of the Ridgecrest earthquake, particularly through kinematic finite-fault inversions (e.g., Goldberg et al., 2020) , although the resulting models show large variability (Wang et al., 2020).

Dynamic rupture simulations (e.g., Day, 1982; Geubelle and Rice, 1995; Day et al., 1998; Barall, 2009; Lapusta and Liu, 2009; Harris et al., 2018) can incorporate physical laws and complex subsurface structures to better constrain earthquake processes. Taufiqurrahman et al. (2023) developed a 3D dynamic rupture model of the M_w 7.1 Ridgecrest earthquake that integrates long- and short-term stress changes, seismic and geodetic data, a 3D velocity model, conjugate fault geometries, topography, and off-fault visco-plasto-elastic deformation (Text S1). Yet, how reliably surface displacement data can constrain on-fault frictional parameters and off-fault inelastic deformation, and the

* Corresponding author.
E-mail addresses: aangniu96@gmail.com (Z. Niu), maximilian.kruse@kit.edu (M. Kruse), mail@linusseelinger.de (L. Seelinger), nico.schliwa@gmx.de (N. Schliwa), heiner.igel@lmu.de (H. Igel), gabriel@geophysik.uni-muenchen.de (A.-A. Gabriel).

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extent of their trade-offs, remains statistically unresolved. In particular, while fault friction parameters are partly constrained by laboratory studies (e.g., Marone, 1998), direct observational constraints on off-fault inelastic properties remain difficult to constrain due to the scarcity of high-resolution, near-fault field observations (e.g., Roten et al., 2014).

Off-fault inelasticity is a key mechanism affecting earthquake physics. Accurately resolving off-fault plasticity and damage properties is essential for modeling near-fault ground motions (e.g., Andrews, 2005; Niu et al., 2025b) and (multi-fault) rupture dynamics (e.g., Finzi and Langer, 2012; Gabriel et al., 2013; Niu et al., 2025a). Off-fault deformation may also help explain the commonly observed deficit in near-surface fault slip, known as 'shallow slip deficit' (SSD, Fialko et al., 2005; Kaneko and Fialko, 2011; Xu et al., 2016; Barnhart et al., 2020; Marchandon et al., 2021; Milliner et al., 2021; Antoine et al., 2024). For the 2019 Ridgecrest mainshock, SSD estimates reach up to 45% (Wang et al., 2020). Multiple mechanisms have been proposed to explain (apparent) SSDs, including post-/interseismic creep, modeling artifacts, and sparse data coverage (e.g., Marchandon et al., 2021). For most earthquakes, including the 2019 Ridgecrest earthquake, these effects may not be large enough to account for the observed SSD (e.g., Antoine et al., 2024). Therefore, co-seismic inelastic off-fault deformation has been proposed as a dominant contributor (e.g., Kaneko et al., 2010; Roten et al., 2017b).

Despite the growing availability of high-quality surface observations, uncertainty quantification using physics-based dynamic rupture simulations remains challenging. One challenge lies in the computational expense, as 3D dynamic rupture simulations of real earthquakes typically require thousands to hundreds of thousands of CPU hours per model realization (e.g., Uphoff et al., 2017; Roten et al., 2014; Gabriel et al., 2023). Dynamic rupture earthquake inversions date back to Mikumo et al. (1987), who used a simplified rupture criterion based on a locking fracture model. Subsequent studies, such as Peyrat and Olsen (2004) and Twardzik et al. (2014), used fully spontaneous dynamic rupture simulations and the neighborhood algorithm (Sambridge, 1999), a nonlinear derivative-free technique, to estimate best-fit models without quantifying uncertainties. For dynamic source inversion with uncertainty analysis (Gallovič et al., 2019), Bayesian inversion based on Markov chain Monte Carlo sampling (MCMC, Hastings, 1970) requires minimal assumptions, however, typically requires thousands to millions of forward simulations (Premus et al., 2022; Schliwa et al., 2024).

Here, we present a Bayesian inversion of geometrically complex 3D dynamic rupture simulations with off-fault plasticity that jointly resolves on- and off-fault nonlinear parameters, using high-resolution satellite imagery and GNSS data. Rather than reducing the computational cost of individual forward simulations, we reduce the number of costly simulations required for uncertainty quantification employing the multilevel delayed acceptance (MLDA) algorithm (Lykkegaard et al., 2023). MLDA efficiently explores the parameter space by combining a hierarchy of fast approximate models with hundreds to thousands of fully resolved runs, correcting for approximation errors. Kruse et al. (2025) recently demonstrated the scalability and efficiency of this approach for large-scale geophysical models in an HPC environment, showing that asynchronous prefetching and surrogate models can provide near-optimal resource allocation in strong-scaling regimes. By leveraging MLDA within the UM-Bridge HPC modeling framework (Seelinger et al., 2023), we efficiently sample posterior distributions using fully resolved SeisSol (Gabriel et al., 2025) dynamic rupture simulations that include nonlinear rheology and realistic 3D subsurface structure. We apply this framework to the 2019 M_w 7.1 Ridgecrest earthquake, performing a Bayesian inversion for rate-and-state fault friction and Drucker–Prager off-fault plasticity parameters. The inversion is constrained by high-rate GNSS time series (Melgar et al., 2020), static GNSS displacements (Floyd et al., 2020), and fault-parallel offsets from satellite image correlation (Antoine et al., 2021). Our results reveal previously unresolved correlations between on-fault friction and off-fault inelastic deformation. We provide the first statistically constrained estimates of fault slip uncertainty and trade-offs with off-fault plasticity from physics-based dynamic rupture inversion. We quantitatively demonstrate that the SSD of the M_w 7.1 Ridgecrest earthquake may be explained by co-seismic off-fault inelasticity. We highlight the potential of probabilistic and physics-based earthquake modeling that integrates observational data with nonlinear rupture dynamics.

2. Data and methods

The objective of this study is to infer nonlinear on-fault and off-fault parameters governing 3D dynamic rupture, constrained by near-fault geodetic observations, and to quantify the associated uncertainties and parameter correlations. While our focus is the well-recorded 2019 Ridgecrest mainshock, our framework is broadly applicable to other earthquakes where high-resolution GNSS and satellite-derived surface deformation data are increasingly available. In this section, we first describe the geodetic datasets that we use to constrain the Bayesian dynamic rupture inversion. We then outline our application of a recently developed, parallelized implementation of the Multilevel Delayed Acceptance (MLDA) algorithm (Kruse et al., 2025), which enables efficient Bayesian inference using computationally intensive dynamic rupture simulations.

2.1. Data

We use fault-parallel offsets from satellite images, high-rate displacement time series from 10 GNSS stations, and static displacements from 11 GNSS stations to constrain the inversion of nonlinear physical parameters in the dynamic rupture source inversion of the 2019 Ridgecrest mainshock. Fault-parallel surface offsets derived from optical image correlation along the modeled fault traces provide key constraints on shallow fault slip distribution. WorldView and Pleiades satellite images offer a ground resolution of 0.5 m and subpixel image correlation achieving displacement resolution down to 0.05 m (Antoine et al., 2021). We define $\Delta u^{obs}(s)$ as the along-strike variation in fault-parallel offset, where s is the distance measured along-strike from the mainshock epicenter (dashed black line in Fig. 1). We note that the fault-parallel surface offsets used here are interpreted products derived from optical correlation and fault-trace projection, and therefore inherit processing and modeling choices (e.g., Barnhart et al. (2020), Antoine et al. (2021), Milliner et al. (2021).

We complement fault-parallel offsets with near-field surface displacement time series from high-rate GNSS stations, to better constrain rupture kinematics. The processed 1 Hz displacement time series from Melgar et al. (2020) have an accuracy of 0.05 m in each displacement component. For later reference, we denote the observed GNSS time series as $u_i^{GT,obs}(t)$, where the index i combines a direction component index j (north-south NS, east-west EW, or up-down UD) and a station index k. Additionally, we incorporate processed co-seismic static displacements from 11 additional GNSS stations provided by UNAVCO, with measurement accuracy between 1.5 and 2.1 mm (Floyd et al., 2020). We denote these observed co-seismic static displacements as $u_i^{GS,obs}$, where the index i again combines the direction component j and the station index k. We independently validate the inversion results against seismic recordings from near-fault strong-motion stations (blue triangles in Fig. 1a; see Section 3.2). We intentionally restrict the observational data set to fault offsets and GNSS (including high-rate GNSS) to keep the likelihood evaluation tractable in a sampling-based inversion with a computationally intensive 3D dynamic rupture model, and to maintain consistency with the observation types used to construct the reference model of Taufiqurrahman et al. (2023).

2.2. Dynamic rupture model parameters

3D earthquake dynamic rupture models combine nonlinear frictional failure along embedded 2D fault surfaces with seismic wave

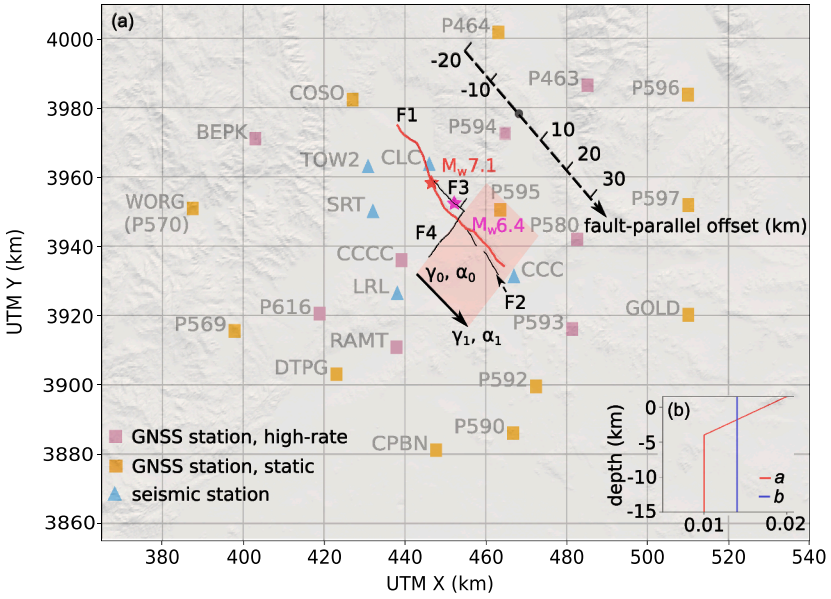


Fig. 1. Data and model setup for the 3D dynamic rupture inversion. (a) Map view of the GNSS and seismic stations used to constrain the inversion, located within 100 km of the M_w 7.1 Ridgecrest mainshock epicenter (red star). The fault trace (F1) of the main fault segment that ruptured during the 2019 M_w7.1 Ridgecrest mainshock is shown as a solid red line, thinner black lines mark the secondary segment ruptured during the mainshock (F2), the subparallel fault (F3) where the M_w6.4 foreshock nucleated (pink star), and the conjugate fault (F4) that hosted the foreshock and re-ruptured during the mainshock, following (Taufiqurrahman et al., 2023). The 3D fault structure is shown in Fig. S2. Pink squares indicate GNSS stations with high-rate displacement time series from Melgar et al. (2020), and orange squares mark GNSS stations used for static co-seismic displacements from Floyd et al. (2020). Blue triangles denote locations of strong-motion seismic stations used to validate the inversion. The region of allowed along-strike spatial variability in off-fault plastic cohesion c and on-fault frictional direct-effect parameter a is highlighted in red shade. Within this zone, c and a vary linearly in space from (γ_0,α_0) to (γ_1, α_1), in the direction of the black arrow. Fault-parallel offset distances are measured along the dashed black axis. (b, Inset) Depth profiles of the reference direct-effect parameter a and the constant evolution-effect parameter b, as used in the strong velocity-weakening friction law (Eq. (6) in SI).

propagation through a 3D elastic or inelastic bulk rock model domain (Harris et al., 2018). The high-resolution reference dynamic rupture model of Taufiqurrahman et al. (2023) provides an important baseline for our inversion. We briefly summarize the rupture dynamics of the reference model in Text S1 and Fig. S1. We intend to infer four model parameters $\mathbf{m} = (\alpha_0, \alpha_1, \gamma_0, \gamma_1)^T$ in this study (Table 1), parameterizing on- and off-fault dynamic rupture parameters.

As in Taufiqurrahman et al. (2023), we prescribe the initial prestress and do not treat it as an inferred parameter. The complex prestress field combines a regional community stress model with Coulomb stress changes from major historical earthquakes (Mw ≥ 7) in the Eastern California Shear Zone over the past ~1,400 years. For discussion of the effects of prestress assumptions on rupture dynamics, we refer to Taufiqurrahman et al. (2023). We fix prestress to keep the inversion computationally tractable and to limit non-uniqueness between prestress and frictional/plasticity parameters. Thus, the inferred posteriors should be interpreted as conditional on the assumed prestress model. Future work may extend the parameterization to jointly infer prestress and constitutive parameters.

2.2.1. *Model resolution and reference rupture dynamics*

Multi-level delayed acceptance (MLDA) is a Markov chain Monte Carlo (MCMC) method that reduces the cost of sampling expensive posteriors by combining several model fidelities (or likelihood approximations) in a hierarchical acceptance scheme. Candidate samples are first screened using cheaper, lower-accuracy models and only a subset is evaluated using the highest-fidelity model, while still targeting the fine-level posterior (see Text S11). For computational reasons, our MLDA algorithm uses two lower-resolution dynamic rupture models within MLDA, both using coarser meshes with a lower resolution compared to the reference model (Section 2.4). The two MLDA models utilize the same mesh, which has ~ 15% of the mesh size of the high-resolution reference model (Text S2). This coarser mesh restricts the resolution of the seismic wavefield to 1.0 Hz close to the fault (instead of 2.0 Hz in the reference model), which remains sufficient to compare with the high-rate GNSS time series used in the inversion. Text S2 documents that the coarser meshes preserve key dynamic rupture characteristics relevant to the inverted data vector. We highlight that minor differences in rupture dynamics due to the coarser mesh do not adversely affect our inversion.

On-fault rupture dynamics are sensitive to on-fault mesh resolution (Day et al., 2005), and the process zone, the area behind the rupture front where shear stress drops from its static value to its dynamic value, must be resolved. In the reference model, the process zone size is >515 m in 95% of all fault elements, and the reference model used 75 m on-fault resolution and polynomial basis functions of order 4. In our model setup, we use a fault element size of 200 m and basis functions of polynomial degree 3 to reduce computational cost. We observe differences in on-fault slip and peak slip rate at the bottom of the seismogenic zone between our high-resolution model and the reference solution (Fig. S4c,d), which are likely due to the complex rupture dynamics leading to deep rupture. In the reference model, a reflected surface rupture front dynamically triggers the deep part of the fault (Schliwa et al., 2025). Increasing resolution is unfortunately out of the scope of current computational capabilities (see Section 2.5). The observed differences in deep rupture dynamics and the resulting small differences in geodetic synthetics fall within the range of uncertainties in our Bayesian likelihood function in Eq. (8), as detailed in Table S1. We discuss differences in rupture dynamics between the high-resolution reference model and our two lower-resolution models in Text S3.

2.2.2. *On-fault friction*

During the inversion, we vary the parameterization of the strong velocity-weakening rate-and-state (SVW-RS) friction law (Dunham et al., 2011), which captures the pronounced reduction in co-seismic friction observed in laboratory experiments at high slip rates (Di Toro et al., 2006). The mathematical formulation of the used SVW-RS

Table 1
Dynamic rupture model parameters, with α_0 and α_1 being scaling factors of the direct-effect parameter a in Eq. (1), L the characteristic slip distance, γ_0 and γ_1 are scaling factors for plastic cohesion c in Eq. (3), and ϕ the bulk friction angle. The on-fault friction parameters that do not vary during the inversion are V_0, f_0, V_w, f_w, V_{init} and b in Eqs. (2) to (7) of the SI.

	Param.	Value	Units	Param.	Value	Units
off-fault	γ_0	[0.1, 2.0]	1	γ_1	[0.1, 2.0]	1
	ϕ	0.7	arc			
on-fault	α_0	[0.4, 1.4]	1	α_1	[0.4, 1.4]	1
	b	0.014	1	L	0.2	m
	f_0	0.6	1	V_0	1.0×10^{-6}	m/s
	f_w	0.1	1	V_w	1.0×10^{-1}	m/s
	V_{init}	1.0×10^{-16}	m/s			

friction law is given in Text S4 and all model parameters are listed in Table 1. Our modeled fault surface extends from the free surface down to $z = -15$ km, and the depth dependent friction parameterization (including $(a-b)$) is defined only over this depth range.

Among the SVW-RS friction parameters, we focus on the physically important parameter combination $(a-b)$, which governs the spontaneity and stability of dynamic rupture (Dunham et al., 2011). Limiting the inversion to a single friction parameter mitigates the high computational cost of each large-scale 3D dynamic rupture simulation, while ensuring accurate posterior probability estimates. The value of $(a-b)$ determines how friction evolves with slip rate, where $a-b<0$ corresponds to velocity-weakening (VW) and $a-b>0$ to velocity-strengthening (VS) behavior. Earthquakes nucleate in VW regions, but fault friction may transition to a VS regime close to the Earth's surface, e.g., due to unconsolidated fault gouge (Kaneko and Fialko, 2011; Cakir et al., 2023). The depth profile of $(a-b)$ plays a critical role in modeling shallow slip behavior during coseismic, postseismic, and interseismic periods (e.g., Barbot et al., 2012).

In our models, we fix $b = 0.014$ and vary a to control the value of $(a-b)$. As a reference, we adopt the depth-dependent profile of $a_0(z)$ from Taufiqurrahman et al. (2023), where z is the upward-positive depth coordinate (Fig. 1b). To parameterize spatial variability in the inversion, we introduce a scalar scaling factor α and define depth-dependent $a(z)$ as:

$$a(z) = \alpha a_0(z). \tag{1}$$

In the fast velocity-weakening formulation of rate-and-state friction (Dunham et al., 2011) that we use, fracture energy is primarily governed by the slip-weakening distance and the effective strength drop, the latter controlled by the frictional parameter combination $a-f_w$. This contrasts with the classical aging and slip laws, where the frictional strength drop is determined by b alone and is independent of a (Rubin and Ampuero, 2005).

2.2.3. Inelastic off-fault deformation and initial stress

The off-fault nonlinear parameters in our model control inelastic deformation in bulk materials during earthquake rupture. We adopt a Drucker-Prager visco-elasto-plastic rheology (Andrews, 2005) to model off-fault inelastic deformation. Inelastic deformation accumulates when the shear stress exceeds a yield strength τ_c, defined as

$$\tau_c = c\cos(\phi) - \sigma_p \sin(\phi), \tag{2}$$

where c is plastic cohesion, ϕ is the bulk friction angle, and $\sigma_p = \sigma_{kk}/3$ is the deviatoric mean normal stress given by the trace of the stress tensor σ_{ij}. The plastic cohesion c and internal friction angle ϕ control the amount of inelastic off-fault deformation and strongly influence the rupture dynamics and the near-field ground motions.

We use the same multi-scale initial stress field as Taufiqurrahman et al. (2023), combining a 2D community stress model representing the regional stress state in the upper crust of Southern California with 3D Coulomb stress changes due to past regional earthquakes. The initial on-fault stress state are shown in Fig. S2 and the regional maximum horizontal stress orientation SH_{max} in Fig. S3. The fault system is embedded in a geologically heterogeneous region, with 3D spatially varying rock elastic moduli based on a 3D community velocity model CVM-S4.26 (Lee et al., 2014). The plastic cohesion $c(x,y,z)$ is proportional to shear modulus $\mu(x,y,z)$ and changes in space according to the 3D community velocity model. We define the plastic cohesion as

$$c = \gamma_r \gamma \mu(x,y,z), \tag{3}$$

where γ is a scaling factor to be inferred during the inversion, and $\gamma_r = 10^{-4}$ (Roten et al., 2014).

In the reference model which uses a spatially uniform on-fault direct-effect scaling parameter $\alpha = 1$ in Eq. (1) and off-fault plastic cohesion scaling factor $\gamma = 1$ in Eq. (3), an overshoot is observed in the southeastern segment of fault F1 when comparing to the surface displacement of GNSS stations and the fault-parallel offset data. To examine whether we can discriminate between on-fault and off-fault nonlinear processes that potentially suppress the overshoot of surface fault offset at the southeastern segment of the fault F1, we assume that the on-fault parameter α and the off-fault parameter γ are both spatially variable along the fault strike direction. We hypothesize that such variability allows for the modeled fault slip in dynamic rupture simulations to closer match observations.

We model along-strike variability by splitting the fault into two regions and assigning separate scaling parameters to the NW and SE segments. For simplicity, we assume that this variation in a and γ along the southeastern segment of fault F1 is linear, and that scaling factors remain constant along the northwestern segment (Fig. 1a). Along the southeastern segment, the scaling factor of the direct-effect parameter a varies from α_0 to α_1; and for the scaling factor of plastic cohesion c from γ_0 to γ_1. We will discuss this assumption in Section 4.1.

2.3. Bayesian inversion of large-scale dynamic rupture simulations

This section describes how we combine data with three different formats in the inversion framework (Section 2.3.1). Then, we formulate the Bayesian inverse problem with detailed explanations on how the covariance matrix in the data likelihood is designed (Section 2.3.2).

2.3.1. Data space

In this study, the data vector comprises two observation types: (i) fault-parallel surface offsets derived from optical correlation and fault-trace projection, $\mathbf{d}^{F,obs}$, and (ii) GNSS observations, $\mathbf{d}^{G,obs}$, including high-rate GNSS time series (and the corresponding static offsets where used), with the forward model predictions evaluated from dynamic rupture simulations. In Section 2.3.2, we partition the observations into fault-offset and GNSS subsets, and introduce $\mathbf{d}^{F,obs} = \left(\mathbf{d}^{NW,obs}, \mathbf{d}^{SE,obs}\right)^T$ and $\mathbf{d}^{G,obs} = \left(\mathbf{d}^{GT,obs}, \mathbf{d}^{GS,obs}\right)^T$ to account for epistemic uncertainties in our dynamic rupture models.

We allow for spatial variation of plastic cohesion c and direct-effect parameter a in the southeastern part of the fault system, via respective scaling factors γ and α (Section 2.2). This results in a higher model flexibility, thus, a lower epistemic uncertainty, in the southeastern part of fault F1. To account for this in the data likelihood function (Section 2.3.2), we separate the fault-parallel offsets from the satellite images $\Delta u^{obs}(s)$ into $\Delta u^{NW,obs}(s)$ for $s \le 10$ km, and $\Delta u^{SE,obs}(s)$ elsewhere, as shown by the dashed black axis in Fig. 1a.

We define this first type of the data for $\Delta u^{NW,obs}(s)$ and $\Delta u^{SE,obs}(s)$, respectively, as $\mathbf{d}^{NW}$ and $\mathbf{d}^{SE}$ in the Bayesian inversion with the following spatial discretization:

$$\mathbf{d}^{k,obs} = d_l^{k,obs} = \Delta u^{k,obs}(s_l),\ k = NW \text{ or } SE, \tag{4}$$

where l runs from 1 to N^{NW} and N^{SE}, respectively, for the northwestern and southeastern parts of the fault. $N^{NW} = 53$ and $N^{SE} = 33$ are the numbers of along-strike discretization points s_l, at which the

northwestern- and southeastern-segment fault-parallel offset values are measured.

Then, we define the second data type, the displacement time series $u_i^{GT}(t)$ at 10 high-rate GNSS stations, as $\mathbf{d}^{GT,obs}$ in the Bayesian inversion with the following temporal discretization:

$$\mathbf{d}^{GT,obs} = d_{il}^{GT,obs} = u_i^{GT,obs}(t_l),\ i = 1, 2, \ldots, n^{GT}, \tag{5}$$

where l runs from 1 to $N^{GT} = 40$, which is the number of discretized time steps t_l with a sampling rate of 1 Hz. The simulation duration is 40 s and $n^{GT} = 3 \times 10$ is the total number of the three-component high-rate displacement time series.

The third type of data consists of the two horizontal components of static surface displacement u_j^{GS} at 11 GNSS stations. We define $\mathbf{d}_j^{GS}$ as

$$\mathbf{d}^{GS,obs} = d_j^{GS,obs} = u_j^{GS,obs},\ j = 1, 2, \ldots, n^{GS}, \tag{6}$$

where $n^{GS} = 2 \times 11$ is the total number of static displacement measurements.

In summary, the complete data vector $\mathbf{d}^{obs}$ that consists of the above three types of surface displacement measurement is defined as

$$\mathbf{d}^{obs} = \left(\mathbf{d}^{NW,obs}, \mathbf{d}^{SE,obs}, \mathbf{d}^{GT,obs}, \mathbf{d}^{GS,obs}\right)^T, \tag{7}$$

with a total dimension of $n_d = N^{NW} + N^{SE} + N^{GT} n^{GT} + n^{GS} = 1308$. Eq. (7) defines a concatenated data vector, but the inversion does not weight all entries equally. As explained in the next Section, relative weighting is controlled by the covariance matrix $\mathbf{C}$. Each residual component is scaled by its assigned variance (and any assumed correlation structure). In particular, GNSS time series enter sample-by-sample. Their aggregate influence (columns "std inv." in Table S1) together with the number of time samples retained.

2.3.2. Data likelihood in Bayesian inversion

In Bayes' theorem, the model parameters $\mathbf{m} = \left(\alpha_0, \alpha_1, \gamma_0, \gamma_1\right)^T$ and the experimental observations $\mathbf{d}^{obs}$ defined in Eq. (7) are treated as random variables in $\mathbb{M} \subset \mathbb{R}^{n_m}$ and $\mathbb{D} \subset \mathbb{R}^{n_d}$, respectively, where n_m and n_d denote the number of model parameters and the number of observed data points. Let $G : \mathbb{M} \to \mathbb{D}$ be the model map that transforms a parameter set into a model prediction. Then, we aim to find the posterior probability density function (PDF), $\rho(\mathbf{m}|\mathbf{d}^{obs})$, which describes the conditional distribution of $\mathbf{m}$ for a given observation $\mathbf{d}^{obs}$. We detail its relation to the model prior density $\rho(\mathbf{m})$ and the data likelihood $\rho(\mathbf{d}^{obs}|\mathbf{m})$ that describes the probability of measuring the observed data given one set of $\mathbf{m}$ in Text S5. The model prior density $\rho(\mathbf{m})$ is assumed to be uniform within the model parameter bounds given in Table 1. In the following, we focus on the definition and formulation of the data likelihood $\rho(\mathbf{d}^{obs}|\mathbf{m})$.

We assume a zero-centered, additive Gaussian noise model for the observed data. It follows that the likelihood $\rho(\mathbf{d}^{obs}|\mathbf{m})$ is given by:

$$\rho(\mathbf{d}^{obs}|\mathbf{m}) \propto \exp\left(-\frac{1}{2}(\mathbf{d}^{obs} - G(\mathbf{m}))^T \mathbf{C}^{-1}(\mathbf{d}^{obs} - G(\mathbf{m}))\right). \tag{8}$$

The covariance matrix $\mathbf{C} \in \mathbb{R}^{n_d \times n_d}$ encapsulates the statistics of the noise, reflecting both measurement precision and epistemic uncertainties due to model assumptions. Given the modest dimensionality of our model space, $\dim(\mathbb{M}) = 4$, we assume that the diagonal components of the covariance matrix $\mathbf{C}$ exceed their respective measurement uncertainties for the fault-parallel offsets, as well as high-rate and static GNSS displacement data.

As described in Section 2.2, the linear spatial variation in model parameters provides greater flexibility for the parameters governing dynamic rupture of the southeastern segment of fault F1. We assume this reduces epistemic uncertainty in that segment. Consequently, we assign different standard deviations for point-wise measurements of the fault-parallel offset, $\sigma_i^{SE} = 0.6$ m for the southeastern segment and $\sigma_j^{NW} = 0.84$ m for the northwestern segment. Since the epistemic uncertainty resulting from the limited model dimensionality ($n_m = 4$) is difficult to quantify precisely, we discuss the sensitivity of the inversion results to the choice of likelihood function in Text S6 and Fig. S5.

Similarly, we assign σ_k^{GT} values ranging from 4 cm to 12 cm for the high-rate GNSS time series (GT), such that the misfit between the reference model and the data is $\approx 2\sigma_k^{GT}$ (Table S1). For the static horizontal GNSS displacements, we set $\sigma_l^{GS} = 1$ cm for each component, which is 5× the reported observed data precision (Floyd et al., 2020). We assume that all observations are statistically independent and define the diagonal covariance matrix in blocks as:

$$\mathbf{C} = \operatorname{diag}\boldsymbol{\Sigma}^{NW}, \boldsymbol{\Sigma}^{SE}, \boldsymbol{\Sigma}^{GT}, \boldsymbol{\Sigma}^{GS}, \tag{9}$$

where $\boldsymbol{\Sigma}^{NW} = \Sigma_i^{NW} = N^{NW}\left(\sigma_i^{NW}\right)^2$, $\boldsymbol{\Sigma}^{SE} = \Sigma_j^{SE} = N^{SE}\left(\sigma_j^{SE}\right)^2$, $\boldsymbol{\Sigma}^{GT} = \Sigma_k^{GT} = N^{GT}\left(\sigma_k^{GT}\right)^2$, and $\boldsymbol{\Sigma}^{GS} = \Sigma_l^{GS} = \left(\sigma_l^{GS}\right)^2$ are four vectors of variances with lengths of N^{NW}, N^{SE}, $N^{GT} n^{GT}$, and n^{GS}, respectively.

The posterior density $\rho(\mathbf{m}|\mathbf{d}^{obs})$ can be computed point-wise up to an unknown constant factor $\rho(\mathbf{d}^{obs})$. Each sample thus requires an evaluation of the model map $G(\mathbf{m})$, i.e., a dynamic rupture simulation. Since each such simulation can be computationally expensive, we employ the MLDA algorithm detailed in the following, which ensures efficient sampling of the posterior using a hierarchy of approximate and high-fidelity simulations.

2.4. Hierarchical Bayesian sampling via MLDA with prefetching

Building on the MCMC sampling method (Hastings, 1970), the MLDA algorithm achieves computational efficiency by combining many fast approximate models with a small number of fully resolved simulations to correct for approximation errors. A major drawback of traditional MCMC methods (e.g., Cui et al., 2016) is their requirement for a large number of forward model evaluations to obtain sufficiently uncorrelated samples of the target posterior density π. To reduce the number of costly evaluations, we employ the MLDA algorithm (Lykkegaard et al., 2023) with a three-level model hierarchy. MLDA employs MCMC sampling on low-cost approximate models to produce high-quality proposals for MCMC sampling steps on more accurate, computationally intensive levels.

Our model hierarchy consists of a Gaussian process regression (GPR, Schulz et al., 2018) surrogate model at the coarsest level $l = 1$. This surrogate model guides the sampling process by proposing mostly decorrelated samples for the finer (higher-resolution) levels ($l = 2, 3$) at very low computational cost. For the results presented in Section 3, we train the level $l = 1$ surrogate model (a statistical emulator) using 87 SeisSol simulations at level $l = 3$ with parameters randomly sampled from the model space $\mathbb{M}$, enabling it to map the input model parameters $\mathbf{m}$ to likelihood estimates in Eq. (8). In other words, $l = 1$ is 'based on' the fine model through supervised training rather than a reduced-resolution physics discretization. The models at levels $l = 2$ and $l = 3$ both use high-resolution SeisSol simulations on computational meshes with approximately 4 million tetrahedral elements (Section 2.2.1). SeisSol's polynomial degree of the basis functions is set to two on level $l = 2$ and three on level $l = 3$. Between two accepted samples at level $l = 3$, we evaluate two samples at level $l = 2$, along with an additional 60 samples at level $l = 1$. This multilevel scheme can reduce the autocorrelation between samples by up to 90% at the finest level ($l = 3$), allowing posterior estimation with drastically reduced numbers of expensive simulations. In addition, we implement a prefetching technique (Angelino et al., 2014) within MLDA to improve scalability on supercomputers. Prefetching enables a parallel speedup of up to four times by simultaneously running multiple candidate simulations within each parallel Markov chain (Kruse et al., 2025). We note that although this study uses a four-parameter model space (where simpler surrogates may suffice), the same surrogate construction is designed to remain applicable as the parameterization is extended to higher dimensions. In addition, the GPR surrogate can be adaptively refined during the MLDA sampling by assimilating newly evaluated fine-model samples (Kruse et al., 2025).

2.5. Computational cost

We perform the inversion using the prefetching MLDA implementation of Kruse et al. (2025), interfaced through UM-Bridge (Seelinger et al., 2023) to control simulation runs on the Frontera supercomputer (Stanzione et al., 2020). Frontera is equipped with Intel Xeon Platinum 8280 ("Cascade Lake") processors, each with 56 cores per node operating at 2.7 GHz. The total number of available compute nodes is 8368. On this machine, SeisSol models at levels $l = 2$ and $l = 3$ require approximately 20 and 10 node hours per simulation, respectively. The GPR surrogate model at level $l = 1$ costs < 1 s in each model evaluation.

For this inversion, we utilized 4096 Frontera nodes for ~19 hours, resulting in a total computational cost of 4,358,144 CPUh. Increasing the resolution of the highest-fidelity forward model would substantially increase the computational demands. For example, increasing the degree of polynomial basis functions from 3 to 4 would increase the computational cost by a factor of 7.

3. Results

We present Bayesian inference results for linearly varying off-fault plastic cohesion (γ_0, γ_1) and on-fault direct-effect parameter in SVW-RS friction (α_0, α_1). We assume independent uniform priors for α_0, α_1, γ_0, γ_1 over the physically plausible bounds listed in Table 1. Within these bounds, the prior does not regularize toward a preferred parameter set, so the posterior structure reflects the constraints imposed by the data (and the chosen model setup). Following Kruse et al. (2025), we optimize the configuration of the MLDA hierarchy to maximize the number of effective posterior samples. Using 16 MLDA chains with two prefetching threads each, and running SeisSol in parallel on 4096 nodes, we generate a total of 912 samples on the finest level. To mitigate initialization bias in estimating the Bayesian posterior $\rho(\mathbf{m}|\mathbf{d}^{obs})$ (Eq. (8) in the SI), we discard the first 20 samples in each Markov chain as "burn-in", based on analysis of the autocorrelation functions (Fig. S6a) and dummyTXdummy-(rank-normalized $\hat{R}$ -statistics (Fig. S6b, Vehtari et al., 2021) of the MLDA chains. After excluding the "burn-in" phase samples, 592 effective samples remain.

Rupture dynamics in all preferred models closely resemble those of the observationally validated reference model (SI Text S1, Movie S1), with some differences including reduced shallow peak slip rates and reduced slip on the secondary fault segment F2 (Fig. S7). In our strongly velocity-weakening regime, and with limited off-fault deformation, coseismic slip is governed primarily by the dynamic stress drop. Thus, because prestress is held fixed throughout the inversion, the modeled slip at depth remains nearly constant across the model space.

3.1. Posterior distributions of on- and off-fault parameters

We estimate the 1D and 2D posterior probability density of the model parameters from the effective MLDA samples (Fig. 2). Models with a posterior probability density $\rho(\mathbf{m}|\mathbf{d}^{obs}) > 0.5\rho_{\max}$ are marked by yellow squares, where $\rho_{\max}$ is the maximum posterior probability density of the best-fit model. In the following, we refer to these as "preferred models".

The preferred models generally fall within the higher posterior-density region of parameter space (darker shading in Fig. 2), rather than forming a distinct cluster beyond what is implied by the posterior density itself. This confirms that the MLDA chains adequately sample the posterior distribution $\rho(\mathbf{m}|\mathbf{d}^{obs})$.

The mean values among the preferred models are $\gamma_0 \approx 0.6$ and $\gamma_1 \approx 0.7$, corresponding to, respectively, $6.0 \times 10^{-5}\mu(x, y, z)$ for the northwestern plastic cohesion and $7.0 \times 10^{-5}\mu(x, y, z)$ for the plastic cohesion around the southeastern ends of the fault system (Fig. 1). These values are consistent with the peaks of the 1D marginal distributions (Fig. 2a, f).

Similarly, the mean on-fault direct-effect parameters are $\alpha_0 \approx 0.8$ in the northwest and $\alpha_1 \approx 1.2$ in the southeast (dark blue region, Fig. 2l). Below 4 km depth, this corresponds to a linear increase in $(a - b)$ from -0.006 to -0.002 along fault F1, indicating spatially varying velocity-weakening behavior with less weakening occurring during dynamic rupture in the southeast. Because the prior is uniform within the bounds of Table 1, the clustering of posterior mass in Fig. 2 reflects information gained from the data under the adopted model assumptions.

The estimated posterior density reveals information on the trade-offs and correlations between off-fault plastic cohesion and on-fault direct-effect parameter in explaining surface displacement observations. In the northwestern F1 segment, the 2D marginal posterior distribution of γ_0 and α_0 shows a clear positive correlation (Fig. 2c), which suggests that locally smaller off-fault inelastic deformation (larger γ_0) can be partially compensated by reduced velocity-weakening frictional behavior (larger α_0). In contrast, no such correlation exists in the southeastern segment of the fault (Fig. 2h). In the southeastern segment, preferred models are concentrated around $\gamma_1 \approx 0.7$ and $\alpha_1 \approx 1.2$, indicating that both increased off-fault inelastic deformation and reduced velocity-weakening friction are required to suppress the local offset overshoot seen in the reference model.

To validate consistency with the surrogate model, we compare the MLDA posterior density estimates to the data likelihood $\rho(\mathbf{d}^{obs}|\mathbf{m})$ computed for the 87 samples (red dots in the sub-diagonal panels of Fig. 2) used to pre-train the GPR model. High-density MLDA posterior regions (highlighted by the dashed red rectangle in Fig. 2c) overlap with the high-likelihood surrogate training points (dots with darker red color in Fig. 2i).

The simple pre-trained surrogate model at level $l = 1$ (color-coded in gray in the sub-diagonal panels of Fig. 2) is allowed to under-represent the probability distribution. For example, the surrogate model misses the local high-probability density region in Fig. 2e (dashed red rectangle) that is recovered by the MLDA inversion (Fig. 2b). This highlights an key benefit of our multilevel approach: approximate models accelerate inference, while the final results retain the accuracy of high-resolution 3D dynamic rupture simulations.

3.2. Surface displacement and uncertainty from preferred models

Models from the MLDA chains provide two types of information on how well the simulations explain observed surface ground displacement. First, similar to deterministic inversion approaches, the preferred models reduce the misfits between the model predictions and the observations. Second, enabled by the Bayesian framework, the ensemble of MLDA samples quantifies uncertainties in the predicted surface displacements.

The preferred models reduce the fault-parallel offset overshoot in the southeastern part of fault F1 ($s = 10 - 30$ km, Fig. 3) by 42% compared to the reference model. The peak offset is reduced from 3.2 m to ~2.4 m (yellow curve). The standard deviation of the modeled fault-parallel offsets ranges from 0.2 to 0.9 m, consistent with the standard deviations $\sigma_i^{SE} = 0.6$ m and $\sigma_j^{NW} = 0.84$ m used in the data likelihood $\rho(\mathbf{d}^{obs}|\mathbf{m})$ in Eq. (8). The largest standard deviations occur between $s = 5$ km and $s = 20$ km, and cover the misfits between the preferred models and the data.

We define the normalized misfits e^{NW} and e^{SE} for the northwestern ($s \leq 10$ km) and southeastern ($s > 10$ km) segments as:

$$e^{NW} = \sqrt{\frac{\left(\mathbf{d}^{NW}(\mathbf{m}) - \mathbf{d}^{NW,obs}\right)^T \left(\mathbf{d}^{NW}(\mathbf{m}) - \mathbf{d}^{NW,obs}\right)}{N^{NW}}}, \tag{10}$$

$$e^{SE} = \sqrt{\frac{\left(\mathbf{d}^{SE}(\mathbf{m}) - \mathbf{d}^{SE,obs}\right)^T \left(\mathbf{d}^{SE}(\mathbf{m}) - \mathbf{d}^{SE,obs}\right)}{N^{SE}}}, \tag{11}$$

where $\mathbf{d}^{NW}(\mathbf{m})$ and $\mathbf{d}^{SE}(\mathbf{m})$ are the modeled fault-parallel offsets for a given parameterization $\mathbf{m}$. Other variables are defined in Eq. (9).

The normalized misfit for the mean preferred models is $e^{NW} = 0.99$ m (Fig. 3). This relatively large misfit may indicate that additional small-scale spatial variability, particularly in plastic cohesion c or the

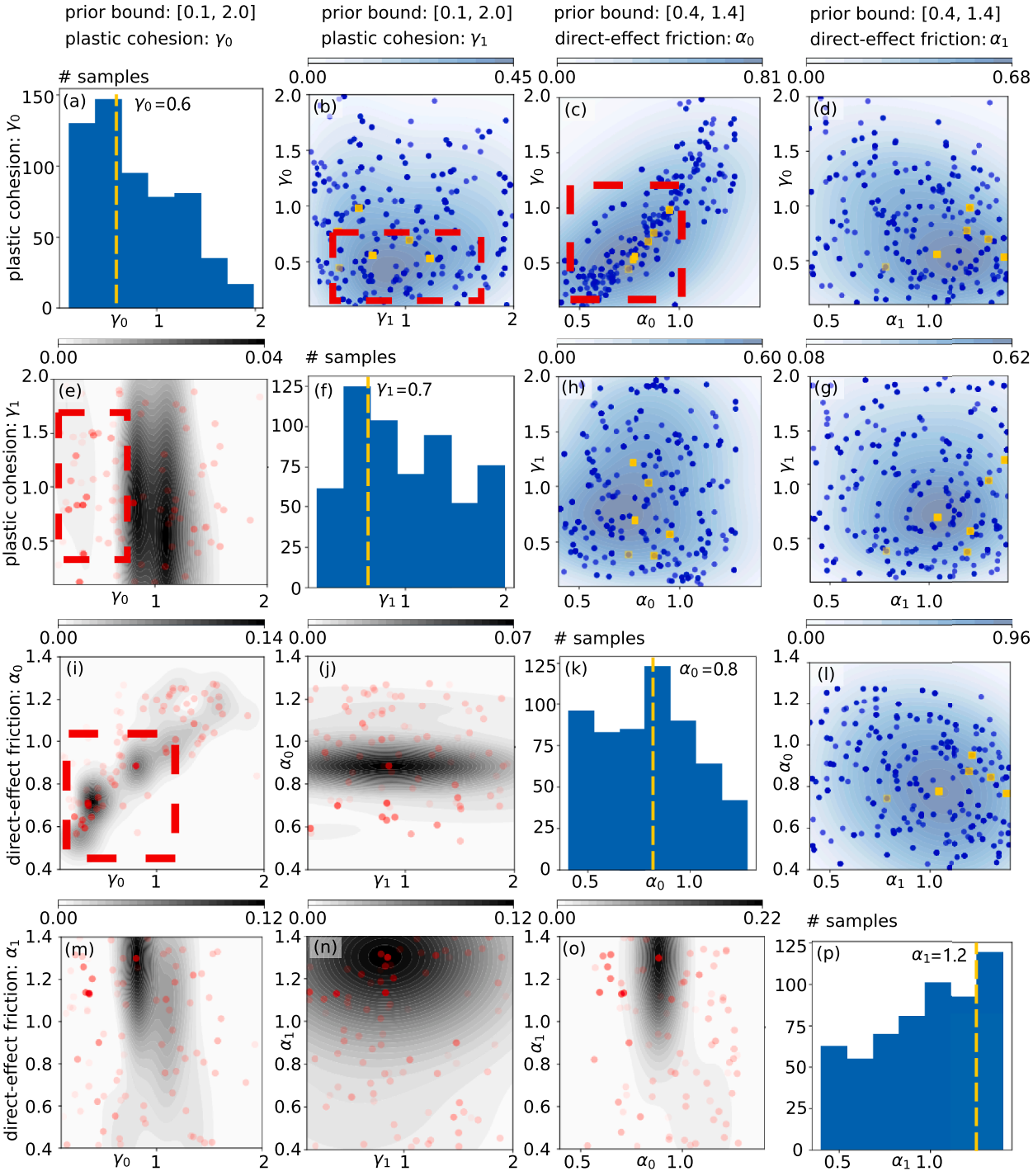


Fig. 2. Panels of the Bayesian likelihood $\rho(\mathbf{d}^{obs}|\mathbf{m})$ from the pretrained surrogate model and the Bayesian posterior $\rho(\mathbf{m}|\mathbf{d}^{obs})$ from the MLDA inversion. The diagonal panels (from top to bottom) show the 1D marginal posterior probabilistic distributions of the off-fault plastic cohesion (γ_0, γ_1) and the on-fault direct-effect parameters in SVW-RS friction (α_0, α_1). The super-diagonal panels show the 2D marginal Bayesian posterior distributions for each parameter pair inferred from the MLDA inversion. The blue dots are all effective samples from the 16 Markov chains. The density of the dots is color-coded in blue, proportional to the posterior $\rho(\mathbf{m}|\mathbf{d}^{obs})$. The density color-code is computed using a Gaussian kernel density estimator (Kristan et al., 2011). The sub-diagonal panels show the 2D marginal Bayesian likelihoods as computed by the surrogate model, color-coded in gray. Red dots are the 87 samples used to pretrain the surrogate model; darker reds correspond to samples with higher likelihood $\rho(\mathbf{d}^{obs}|\mathbf{m})$. We assume independent uniform priors for α_0, α_1,γ_0, γ_1 within the bounds listed in 1. Since it is constant on that support, it is not plotted separately in these posterior panels. .

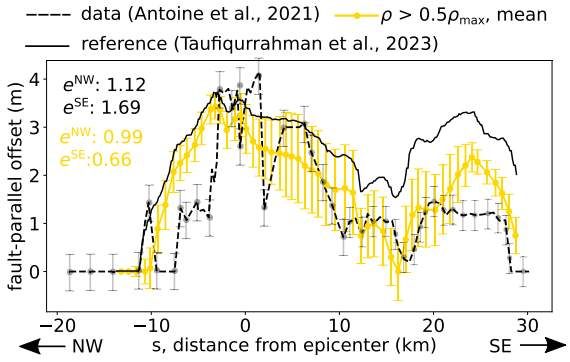


Fig. 3. Fault-parallel offsets from model parameterizations with high posterior probability density. The yellow curve shows the mean offsets of the models with posterior probability density $\rho(\mathbf{m}|\mathbf{d}^{obs}) \geq 0.5\rho_{\max}$. $\rho_{\max}$ is the maximum posterior density $\rho(\mathbf{m}|\mathbf{d}^{obs})$ among all models from the 16 MLDA chains. The yellow error bars show the standard deviation of the modeled offsets among all MLDA models. The dashed black curve shows observed fault-parallel offset data from Antoine et al. (2021), while the gray error bars show the $\sigma_i^{NW} = 0.84$ m and $\sigma_i^{SE} = 0.6$ m in the likelihood, Eq. (8). The solid black curve shows the modeled offsets from the reference dynamic rupture model of Taufiqurrahman et al. (2023) in the solid black curve. The numbers are the misfits between the data and our preferred modeled offsets in the northwest (e^{NW}) and southeast (e^{SE}) of the fault in meters, as defined in Eqs. (10) and (11). The colors of the numbers correspond to those of the solid curves.

direct-effect parameter a, could improve the fit. The large standard deviations in this region are consistent with this interpretation. Along the southern segment ($s = -10$ km to $s = 5$ km), however, the offsets show persistent data misfits not fully explained by variations in c and a. We imposed a linear variation of parameters to counter the overshoot of the reference model, but, because α_1 and γ_1 describe only a limited part of the fault, they may contribute little weight to the misfit function, which in turn may explain the lack of trade-off observed between these parameters.

Among the 21 GNSS stations used in our inversion, the observed static displacements fall within two standard deviations of the model predictions (Fig. 4, blue filled circles) at 14 of the 21 stations in the preferred models. This corresponds to a 45% reduction in misfit compared to the reference model. Specifically, our preferred models reduce the static displacement at most GNSS stations south of the faults (e.g., stations DTPG, P592, CPBN, Fig. 4). This reduction is due to the ≈ 1.3 m smaller cumulative fault slip and the smaller rupture area in our preferred models (Fig. 5a) compared to the reference model (Fig. S4a). Similarly to the fault-parallel offsets, discrepancies in displacements not covered by the model uncertainties indicate that a higher-dimensional parameter space may be required to match the spatial complexity of the rupture.

The preferred models (yellow dots in Fig. 2) match the observed seismic waveforms better than the reference model, primarily due to a ~30% reduction in slip rate within the top 2km of the fault (Movie S1) due to enhanced off-fault plastic yielding (Figs. S11, S12). For example, they reduce the velocity amplitudes at the seismic station CCC by ~55% (Fig. S9). While the moment release rate of the preferred models retains the same mean moment magnitude ($M_w = 6.98 \pm 0.01$) as the reference model, the second peak is lower (Fig. S10), in agreement with kinematic inversions (e.g., Goldberg et al., 2020).

3.3. Uncertainty in fault slip distribution and shallow slip deficit from dynamic rupture inversion

The ensemble of models from the MLDA chains allows us to derive the preferred fault slip distribution on fault F1 as well as to quantify its uncertainty. Fig. 5a shows the mean cumulative slip from the preferred models, while Fig. 5b shows the corresponding standard deviation (std). A small fault segment (blue line in Fig. 5a), where the rupture does not reach the surface in the preferred models, corresponds to a location of incomplete surface rupture. The local prestress ratio (defined in Eq. (1) in the SI) is as low as 0.2 (Fig. S2b), reflecting the unfavorable orientation of this fault segment relative to the background stress. Unlike the reference model, the enhanced off-fault inelastic deformation dissipates enough energy to inhibit local rupture to the surface. This local lack of surface rupture is consistent with previous kinematic inversion results (Antoine et al., 2024). The largest fault slip uncertainties occur at shallow depths, within approximately ~3 km of the surface, where standard deviations exceed 0.4 m. This is the depth range where on-fault friction transitions from velocity-weakening to velocity-strengthening (Fig. 1b), and where inelastic deformation becomes prominent, due to the shallowly reduced shear modulus (Fig. S11). Because plastic cohesion is assumed to decrease linearly with shear modulus, inelastic strain begins to accumulate at lower dynamic stress levels. Within this depth range, the fault slip between $s = 5 - 20$ km along-strike exhibits the largest uncertainties, with standard deviations exceeding 0.7 m. These larger uncertainties align with large variations in fault-parallel offsets within the same along-strike region (Fig. 3). In contrast, the fault slip between $s = 20$ km and $s = 30$ km along-strike shows lower uncertainties, with standard deviations between 0.3 m and 0.4 m. Here, the variability reflects ambiguities in the location of rupture termination (blue circles, Fig. 5b). While the preferred models suggest rupture over a limited area of the fault system, the MLDA sampling does not exclude models with larger rupture extents.

Uncertainties in the predicted slip and near-fault deformation concentrate in the shallow crust, where off-fault inelasticity and frictional transitions exert strong control on rupture dynamics. This differs from static (geodetic) inversions, in which uncertainty often grows with depth due to diminishing sensitivity. At greater depths, the relatively small posterior slip variability in our models should therefore be interpreted as a consequence of the adopted physics-based constraints, which restrict the admissible solution space. The contrasting depth dependence reflects the nonlinear mapping from the parameter prior to slip through dynamic rupture physics, together with the depth-dependence of shear modulus. In our models, plastic cohesion is assumed as proportional to shear modulus (Roten et al., 2014). In addition, under strong velocity-weakening friction, rupture dynamics can change abruptly depending on whether the characteristic weakening velocity is reached (Ampuero and Ben-Zion, 2008; Gabriel et al., 2012). Consequently, at shallow depth, small changes in on-fault or off-fault properties can produce large changes in slip. In this sense, the dynamic-rupture inversion is complementary to geodetic slip inversions, because it propagates uncertainty in the shallow dynamic rupture physics (off-fault yielding and near-surface frictional dynamics), where surface geodetic observations have the largest sensitivity.

We use the MLDA ensemble to statistically evaluate the SSD from our physics-based dynamic rupture models (Fig. 5a). The SSD is commonly defined as the percentage of one minus the ratio of average fault slip at the surface to the maximum fault slip at depth (Fialko et al., 2005). As shown in Fig. 5c, more than 65% of the models exhibit low SSDs around a peak of ~10%. The mean SSD among the preferred models is 13.1% (yellow curve and error bars in Fig. 5d) with a standard deviation of 5.1%. We observe that fault slip uncertainty increases toward shallower depths, consistent with the greater variability in modeled near-surface deformation.

4. Discussion

4.1. Trade-offs between off-fault plasticity and frictional weakening and implications for fault maturity and fault zone structure

Off-fault plastic cohesion (c, Eq. (2)) and on-fault rate-and-state friction parameters ($a - b$, Eq. (3) in the SI) jointly affect rupture dynamics and the resulting fault slip distribution and off-fault deformation (e.g.,

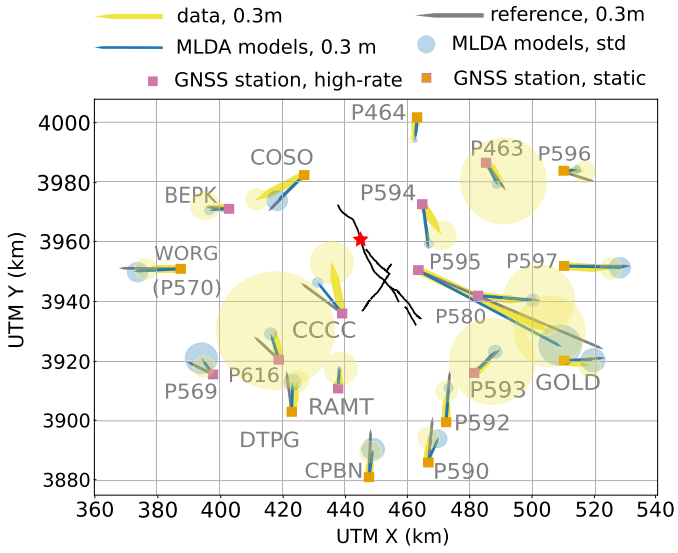


Fig. 4. Co-seismic horizontal static displacements at 21 GNSS stations. Orange squares indicate stations used only for static displacement data, and pink squares mark those stations where high-rate displacement observations are available. Blue arrows show the mean modeled displacements from the preferred models, with circle radii equal to two times the standard deviation from the MLDA ensemble. Yellow arrows denote static GNSS displacement data from Floyd et al. (2020), with circle radii equal to $2\sigma_k^{GT}$ and $2\sigma_l^{GS}$ in the likelihood, Eq. (8). We provide the detailed values in Table S1 of the SI. The gray arrows denote results from the reference dynamic rupture model (Taufiqurrahman et al., 2023). The modeled and observed 40 s displacement time series at 10 high-rate stations are provided in Fig. S8.

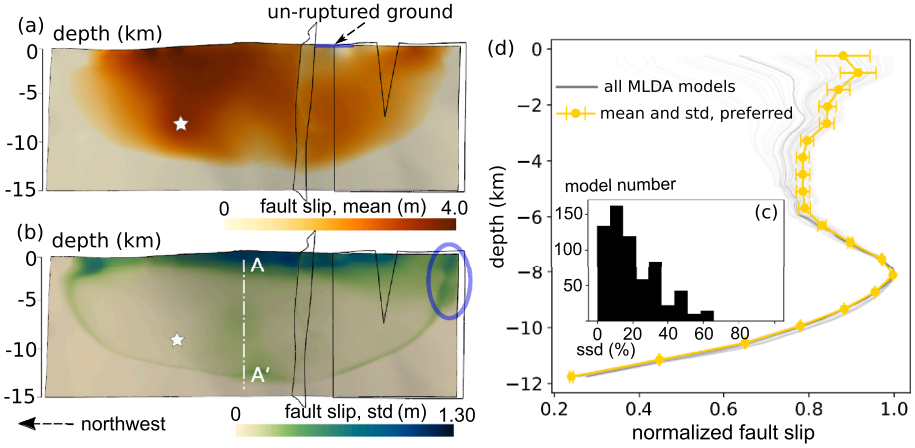


Fig. 5. Uncertainties in fault slip distribution among all models in the MLDA chains. Panels (a) and (b) show, respectively, the mean and the standard deviation of fault slip. White stars mark the hypocenter where the dynamic rupture models prescribe earthquake nucleation. (c) Histogram of shallow slip deficits (SSDs) across all models. (d) Variation of fault slip with depth. The yellow curve shows the mean fault slip among the preferred models (yellow dots, Fig. 2), error bars show depth-dependent fault slip standard deviations. Gray curves show results from all accepted models. The off-fault inelastic deformation around the A-A' cross-section in (b) is shown in Fig. S11. Because the prior is imposed on **m** (Table 1) rather than directly on slip, the implied prior predictive variability of slip is depth-dependent through the nonlinear rupture physics and depth-varying material and frictional parameters.

Biemiller et al., 2023). Our Bayesian inversion reveals trade-offs between off-fault plastic cohesion and on-fault frictional weakening specifically in the northwestern part of the fault system. This indicates that locally co-seismic surface deformation may be equally well explained by different combinations of inelastic and frictional properties, which reflects the competing roles of off-fault damage and fault weakening in dissipating earthquake energy. Plastic cohesion controls energy dissipation through off-fault inelastic deformation, competing with the energy loss from frictional heating in consuming the total energy released from the earthquake (Nielsen et al., 2016). On the other hand, $(a-b)$ influences the amount of energy dissipation during unstable fault slip (Dunham et al., 2011). The Bayesian inversion with MLDA enables us to investigate whether (and how) it is possible to discriminate the effects of c and $(a-b)$ on the on- and off-fault deformation from observed surface displacements, providing physical insights that extend beyond best-fit models.

In the southeastern part of the fault system, matching the observed fault-parallel offsets requires both enhanced inelastic deformation and reduced frictional weakening, suggesting that neither mechanism alone is sufficient. These spatial variations in the interplay of on-fault friction and off-fault damage may reflect along-strike variations in fault zone maturity. A supporting observation is the relatively

discontinuous surface fault trace mapped along the southeastern part of fault F1 prior to the 2019 Ridgecrest earthquake (Thompson Jobe et al., 2020), which may suggest a less mature fault zone (Dolan and Haravitch, 2014; Manighetti et al., 2021), which is typically associated with less severe velocity-weakening (Ikari et al., 2011).

The reduced plastic cohesion favored by the inversion to match surface deformation may reflect the presence of a shallow fault zone surrounding the ruptured fault segments (e.g., Roten et al., 2017a). In the high-probability subset of the inversion (yellow squares, Fig. 2), lower plastic cohesion c is achieved by decreasing the scaling factor γ that relates it to the shear modulus $\mu(x, y, z)$ (Eq. (3)). Imaging studies support this interpretation, having identified shallow fault zones, 3–5 km deep, surrounding the Ridgecrest rupture area (Qiu et al., 2021; Zhou et al., 2022), which considerably affects rupture dynamics and the resulting ground shaking (Yeh and Olsen, 2023; Schliwa et al., 2025). For example, Qiu et al. (2021) report a shallow waveguide with a 20% reduction in shear wave velocity relative to the surrounding host rock, corresponding to a ~36% reduction in shear modulus, consistent with our preferred model values of $\gamma_0 \approx 0.6$ and $\gamma_1 \approx 0.7$.

4.2. Uncertainty in shallow slip deficit and the role of inelastic deformation

We quantitatively compare the SSD inferred from our ensemble of dynamic rupture simulations with estimates from kinematic fault slip inversions (Wang et al., 2020), to assess whether observed SSDs can be reproduced. In the southeastern part of the fault system, lower plastic cohesion (γ_0 and γ_1) and less prominent velocity-weakening behavior (larger α_1) reduce the fault-parallel offset in our preferred models (yellow curve, Fig. 3). This reduced cohesion leads to more distributed off-fault inelastic deformation (Figs. S11, S12), which in turn increases the SSD.

Estimates of SSD from kinematic inversions range from 2% to 45%, reflecting large uncertainty (Ross et al., 2019; Goldberg et al., 2020; Wang et al., 2020; Antoine et al., 2024). It has been shown that assuming linear elastic rocks in kinematic inversions can overestimate SSD by 5%-25% (Ragon and Simons, 2021; Marchandon et al., 2021), suggesting that a more realistic range of SSD is between 8% and 32% (Antoine et al., 2024). From our Bayesian inversion, ~55% of the MLDA samples fall within this SSD range, demonstrating that co-seismic off-fault deformation can quantitatively account for the observed SSD.

4.3. Toward higher-dimensional inversions

The primary limitation of our inversion is the limited dimensionality of the model space ($n_m = 4$). As reflected in the choice of likelihood covariance matrix (Eq. (8)) and the misfits to the GNSS displacement time series (Fig. S8), this low dimensional parameterization increases epistemic uncertainties and limits the model's ability to match all observations within their measurement accuracies.

The uncertainty analysis suggests that smaller-scale variations in on-fault friction and off-fault plasticity, beyond the here imposed linear along-strike gradients, may improve to fit to observed fault offsets within measurement accuracy (0.05 m), particularly in the range $s = 5 - 20$ km. For $s < 5$km, however, we show that variations in c and a alone are insufficient, implying that additional complexity or physical parameters, especially at depths shallower than ~4 km, may be required. Addressing these limitations would require higher model dimensions n_m to include more parameters with enhanced spatial flexibility at much higher computational costs (Section 2.5).

Our results show that increasing model dimensions at depths deeper than ~ 4 km could help to reduce residual misfits in GNSS data. Unlike the offset data, the GNSS displacement standard deviations in the likelihood function (Eq. (8)) are much smaller, on the order of a few centimeters, and closer to measurement accuracy. GNSS displacements are particularly sensitive to the deeper portions of the fault slip distribution (Antoine et al., 2024). Yet, in our preferred models, residuals at 7 stations (e.g., CCCC) remain outside the two standard deviation uncertainty range. Furthermore, the posterior standard deviations in fault slip at depths below ~ 4 km are relatively small (mostly <0.1 m), implying that further improvements may require incorporating other physical parameters beyond c and a at depth.

Together, these findings suggest that combining the currently available geodetic data with high-performance computing, it may be feasible to invert for a higher-dimensional parameter space-including depth-dependent variations in a and c, and possibly additional parameters governing on- and off-fault rheology.

Generalizing our MLDA inversion framework to higher-dimensional models will require improvements in two aspects. First, improved surrogate models are needed at the coarsest level, capable of generalizing to broader parameter spaces and supporting more complex dynamic rupture scenarios. Promising approaches for dynamic rupture simulations include reduced-order models, such as interpolated proper orthogonal decomposition (Hobson et al., 2025), deep-learning-based methods, such as neural networks (Gong et al., 2025), Fourier neural networks (Tainpakdipat et al., 2025), and derivative-informed neural operators (O'Leary-Roseberry et al., 2024). In future work, directly integrating full-field surface displacement data from InSAR or optical correlation may reduce parameter trade-offs by adding spatial sensitivity to shallow slip gradients and off-fault deformation, but requires explicit treatment of spatially correlated noise and increased computational cost. Second, the number of required costly fine-level dynamic rupture simulations must remain tractable, ideally in the range of a few thousand. This can be achieved through function-space Bayesian inference frameworks (e.g., Stuart, 2010) or by leveraging accurate parametric derivatives computed via adjoint dynamic rupture simulations (Stiernström et al., 2024). However, adjoint or automatic-differentiation gradients are not available for our model setup, coupling strong velocity-weakening rate-and-state friction with off-fault Drucker–Prager visco-elasto-plasticity, so Gauss-Newton (gradient-informed) optimization and local Gaussian posterior approximations are not straightforward to apply or validate. Nevertheless the probabilistic approach allows us to quantify correlations between model parameters, an important contribution to the final interpretation of the physical properties of the system.

5. Conclusions

For the first time, this study demonstrates the feasibility of probabilistic, physics-based earthquake source inversion by integrating 3D dynamic rupture simulations including off-fault inelastic deformation with hierarchical Bayesian inference. Using the multilevel delayed acceptance Markov chain Monte Carlo (MLDA) method, accelerated with prefetching and surrogate models, we achieve efficient uncertainty quantification with costly dynamic rupture simulations. Applied to the 2019 Mw 7.1 Ridgecrest earthquake, our approach combines high-resolution geodetic datasets-fault-parallel offsets from satellite imagery, high-rate GNSS time series, and static GNSS displacements, to constrain coupled nonlinear processes of on-fault frictional weakening and off-fault inelastic deformation.

The inferred posterior distributions reveal strong trade-offs between on-fault frictional weakening (direct-effect parameter, a) and off-fault plasticity (plastic cohesion, (c)). In the northwest of the fault system, lower plastic cohesion can compensate for stronger velocity-weakening behavior, jointly controlling the rupture dynamics and surface deformation. The preferred models favor spatially variable dynamic weakening, with (a-b) increasing from -0.006 in the northwest to -0.002 in the southeast, consistent with geological evidence for lower fault maturity in the southeastern segment. A corresponding gradient in plastic cohesion from $\gamma_0 \approx 0.6$ to $\gamma_1 \approx 0.7$ improves agreement with observed surface displacements and is consistent with seismic imaging of shallow damage zones.

Our Bayesian inversion constrains the shallow slip deficit (SSD) to 13.1% ±5.1% among the preferred models. Notably, 55% of all

MLDA samples fall within the 8–32% SSD range estimated by previous kinematic inversions when accounting for off-fault deformation. This demonstrates that co-seismic inelastic processes are important to quantitatively explain the observed SSD. Our results provide new constraints on the maturity and shallow structure of the Ridgecrest fault system, and illustrate how joint inversion of on- and off-fault processes can improve our physical understanding of rupture mechanics.

This study establishes an openly available HPC-empowered and physics-based framework for probabilistic earthquake source modeling that captures coupled nonlinearities in rupture dynamics and off-fault deformation. Our results pave the way for systematic exploration of fault zone properties, uncertainty quantification, and physics-based seismic hazard assessment.

CRediT authorship contribution statement

Zihua Niu: Writing – original draft; **Maximilian Kruse:** Writing – review & editing; **Linus Seelinger:** Writing – review & editing; **Nico Schliwa:** Writing – review & editing; **Heiner Igel:** Writing – review & editing; **Alice-Agnes Gabriel:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The MLDA setup, simulation outputs, and post-processing scripts for reproducing the results are provided in Niu et al. (2025c). The GNSS data are available from Melgar et al. (2020) and Floyd et al. (2020), and the fault-parallel offset data are available from Antoine et al. (2021).

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.epsl.2026.120061.

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